

MARKETING ANALYTICS PROJECT

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INTRODUCTION

This project presents a comprehensive and quantitative approach to marketing, aiming to improve customer loyalty and engagement using historical data. By integrating **churn predictive models**, **RFM segmentation**, and **sentiment analysis**, we seek to understand customer behavior, identify at-risk segments, and extract actionable insights from feedback.

The analysis leverages rich **datasets via Snowit**, including orders and transactions (frequency, monetary value, product choice, promotions), user profiles (demographics and preferences), loyalty and affiliation data (engagement and status), and customer reviews to assess sentiment and satisfaction levels.

Following the **CRISP-DM methodology**, the project ensures a rigorous process from data collection and preparation, to modeling, and implementation. The ultimate goal is to **design targeted campaigns** that optimize marketing resources, increase loyalty, and deliver measurable business value.

BUSINESS SCOPE

This study aims to explore the following research questions to support data-driven marketing strategies:

- ❖ **How can the average customer value be increased?**
- ❖ **How can seasonal churn be reduced?**
- ❖ **How can we identify and address negative customer experiences to improve overall satisfaction and prevent abandonment?**

These questions form the foundation of the analysis, guiding data collection, modeling, and eventual marketing interventions.

USE CASES

Based on the research questions defined in the Business Scope, the analysis supports the following practical marketing applications:

- ❖ **VIP / Premium Campaigns:** Segment high-value customers and design exclusive promotions to maximize loyalty and lifetime value.
- ❖ **Retention Campaigns:** Identify users at risk of leaving and implement targeted retention strategies to reduce churn.
- ❖ **Recovery & Engagement Actions:** Detect dissatisfied customers and initiate actions to recover satisfaction and strengthen engagement.

This approach enables a proactive, data-driven marketing strategy that improves retention, value, and customer experience.

DATA COLLECTION

The analysis relies on multiple datasets provided by Snowlt, including:

- ❖ **USERS** (Customer information): ID, account details, referrals, city, language, marketing source, preferred locations.
- ❖ **PROFILES** (Profile attributes): birthday, gender, height/weight, ski level, services allowed.
- ❖ **CARDS** (Ski pass card information): status, validity, membership type, assignment details.
- ❖ **ORDERS & ORDER_DETAILS** (Purchase records): items, price, discounts, product type, zone, device info, payment info.

Review Data for Sentiment Analysis

- ❖ **LABELLED_REVIEWS**: Dataset with ground truth sentiment labels (Positive / Negative / Neutral).
- ❖ **REVIEWS**: Unlabelled textual reviews (public, privacy-compliant).

DATA PREPARATION

Regarding data types, we standardized each column to the correct type based on its content: dates in datetime format and booleans as True/False.

We applied different strategies to handle missing values (NaN):

- ❖ **Removal of critical rows:** when essential information for the analysis was missing, the row was deleted since it could not be used reliably.
- ❖ **Replacement with placeholder values or median:** for missing categorical columns, a generic value indicating the absence was inserted; for numerical columns, missing values were replaced with the median of available values, thus preserving the overall distribution.
- ❖ **Removal of almost empty columns:** columns with a very high percentage of missing values were eliminated because they were too sparse to provide meaningful information.

DATA EXPLORATION

Table USERS

Main KPIs:

- ❖ Total Users: 715,839
- ❖ Anonymous Users: 14.5%
- ❖ Users by Source:
 - Website: 65.6%
 - App: 22.2%
 - Third-party: 10%

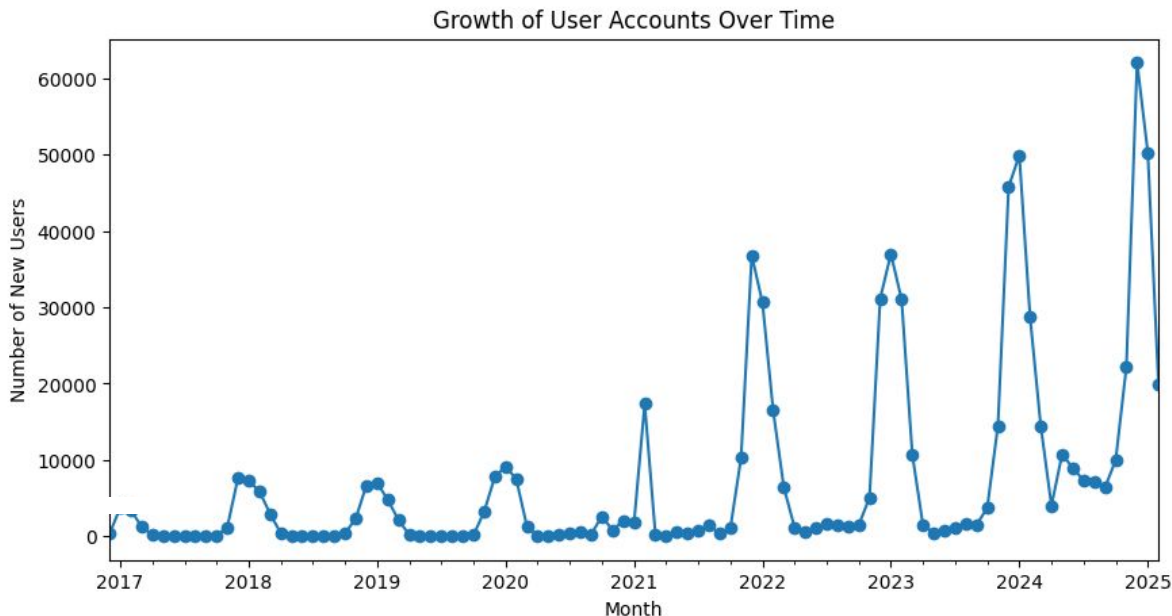


Table PROFILES

Main KPIs:

- ❖ Total Profiles: 97,158
- ❖ Users with >1 Profile: 31.9%
- ❖ Average Age: 27.8 years

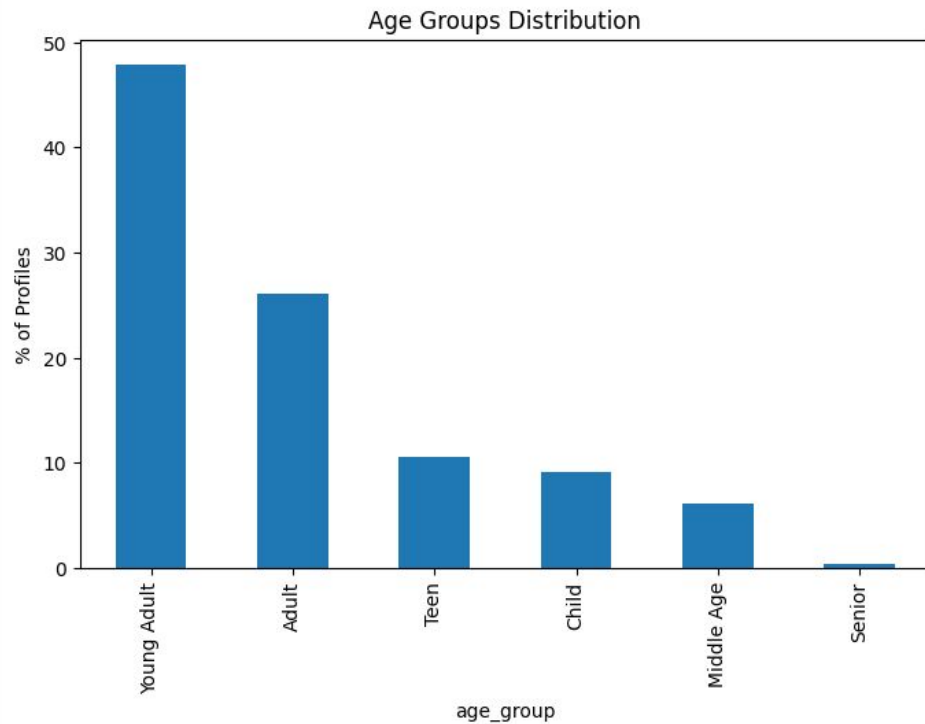


Table CARDS

Main KPIs:

- ❖ Total Cards: 250,869
- ❖ Users with >1 Card: 53.1%
- ❖ Average Cards per User: 2.09

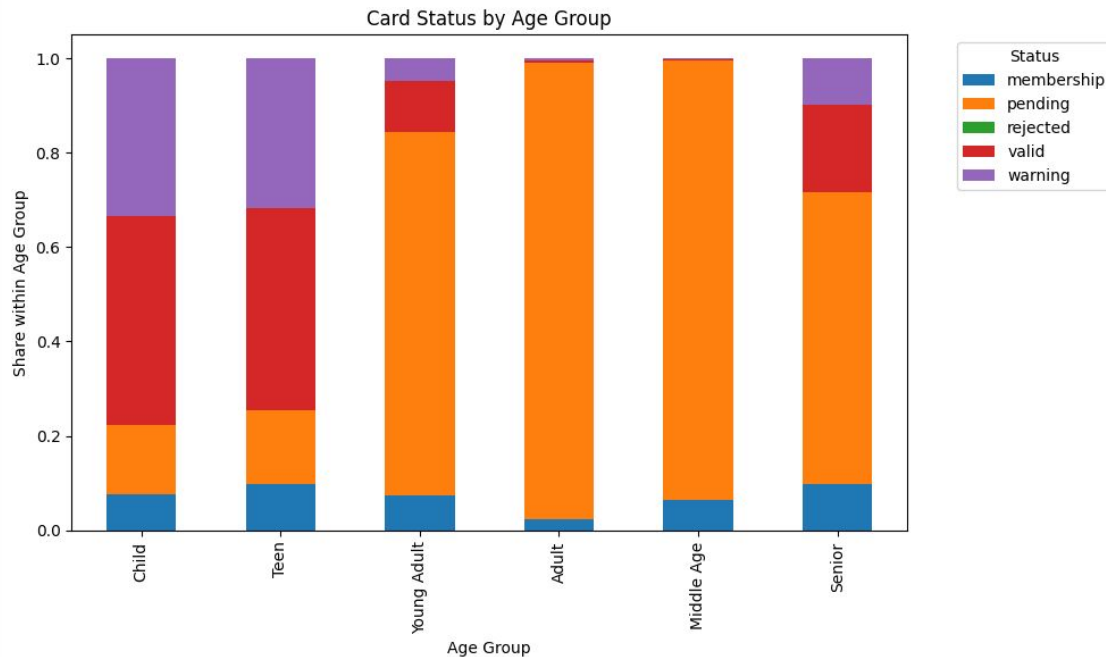


Table ORDERS

Main KPIs:

- ❖ Total Orders: 549,105
- ❖ Repeat Customer Rate: 41.6%
- ❖ Average Orders per User: 2.37

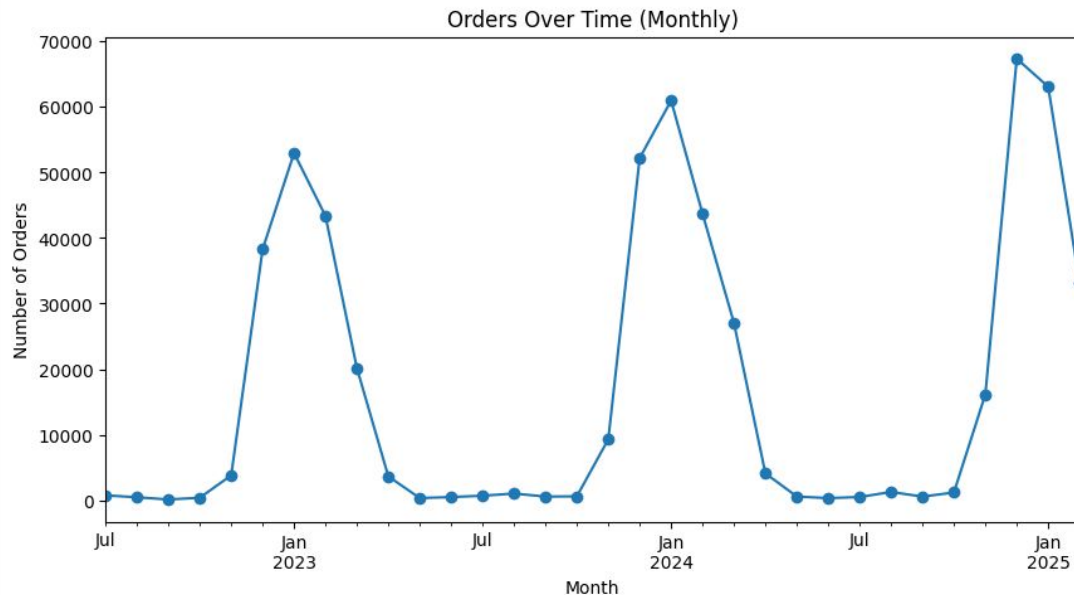
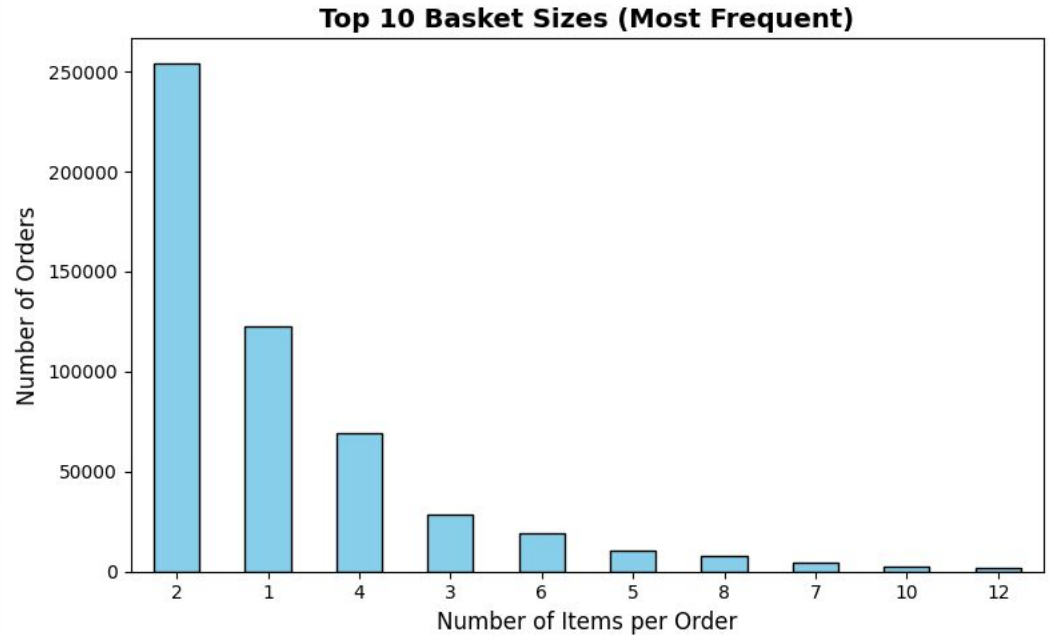


Table ORDER_DETAILS

Main KPIs:

- ❖ Total Items Sold: 1,371,543
- ❖ Total Sales Amount: €56,980,420
- ❖ Average Amount per Item: €41.54



MODELLING

RFM segmentation

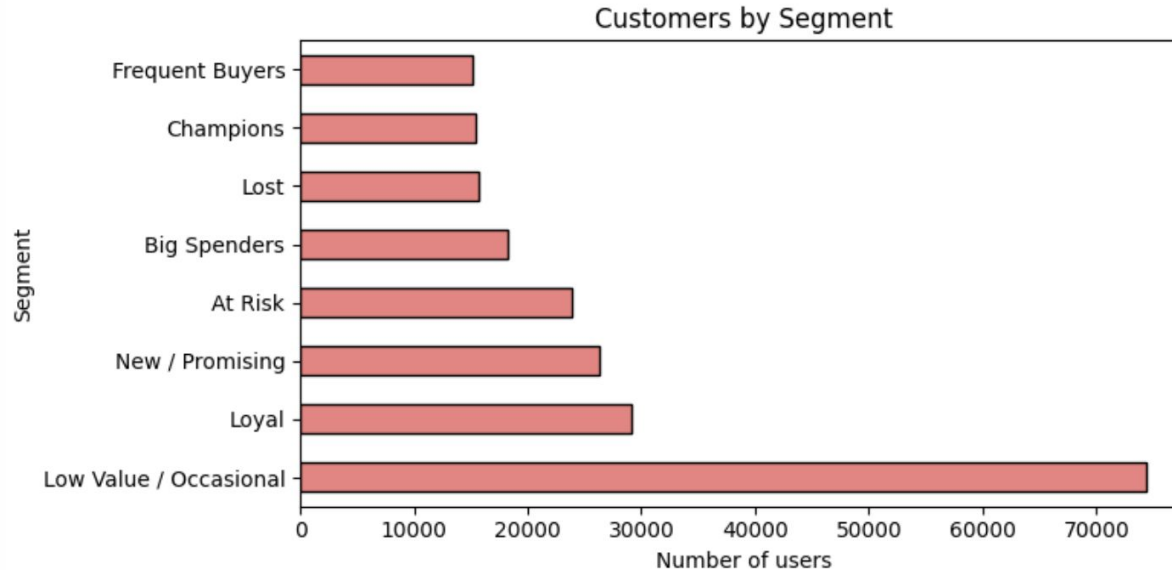
To uncover meaningful customer insights each customer was profiled using:

- ❖ Recency: days since the last order, measuring freshness of engagement.
- ❖ Frequency: total number of orders, reflecting loyalty and activity.
- ❖ Monetary: total spending, capturing the financial contribution of the customer.

All metrics were standardized into **quartile-based scores (1–4)** and combined into an RFM code.

This procedure allowed us to systematically classify customers into segments such as **Champions**, **Loyal**, **At Risk**, **Lost**, and others.

RFM segmentation



This uneven distribution highlights the importance of personalizing marketing initiatives: while most customers require activation strategies, a smaller subset warrants exclusive benefits and premium engagement.

CHURN Prediction Model

To predict user churn on the Snowit platform, we focused only on active users from the previous season, ensuring the model is trained on relevant data using the following steps:

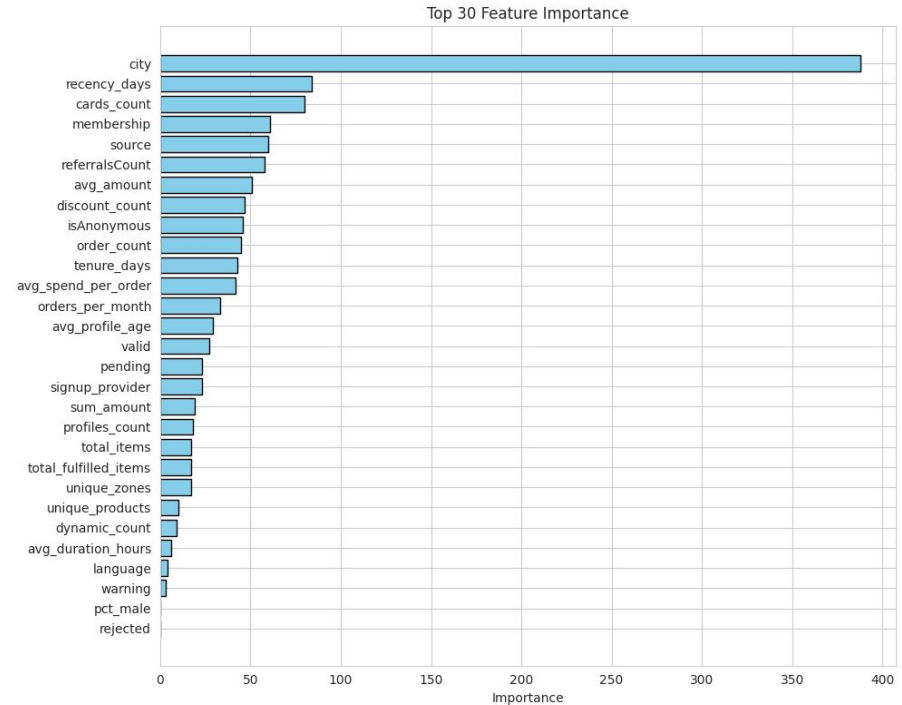
- ❖ **Data cleaning and standardization:**
 - Column names are stripped and standardized across multiple datasets.
 - Datetime columns are analyzed to ensure accurate time-based calculations.
- ❖ **Feature Engineering from Orders:**
 - Filter orders up to the reference date (end of the previous season).
 - Aggregation of order details by user: total items, fulfilled items, discounts ecc.
 - Calculation of user behavior metrics: recency, duration, and orders per month.
- ❖ **Profile and card features:**
 - Profiles: average age, gender distribution, number of profiles.
 - Cards: total assigned, days since assignment, number of statuses.

The goal was to transform raw transactional and user data into structured features that describe user engagement and behavior.

CHURN Prediction Model

We tested several models and found that **LightGBM delivered the best performance.**

- ❖ **Model Performance:**
LightGBM with undersampling achieved strong predictive ability (**AUC 0.82**), ensuring reliable churn identification.
- ❖ **Key Drivers of Churn:**
Key churn drivers are user's **city**, time since last purchase, number of cards, membership type, and acquisition channel.
- ❖ **Business Impact:**
The model allows to **identify at-risk users** and design targeted retention campaigns.



SENTIMENT analysis

This analysis is useful to analyze customer reviews and social media mentions to understand user perception of the Snowit platform.

❖ Data Preparation:

- Text cleaning: removal of stopwords, special characters.
- Standardization: lowercasing, tokenization, and lemmatization.

❖ Feature Engineering:

- Creation of n-grams and TF-IDF representations.
- Sentiment scoring with pre-trained lexicons and ML models.

❖ Modeling:

- Tested multiple classifiers (Logistic Regression, Random Forest and CNN).
- Selected the best-performing approach for polarity detection.

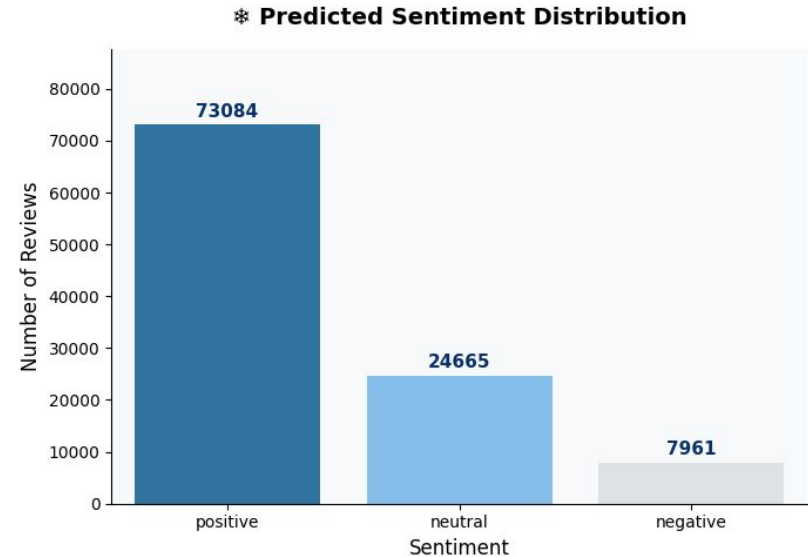
SENTIMENT analysis

Model Performance

The chosen model (LogisticRegression) achieved overall strong **accuracy (0.82)** and **precision** in classifying reviews as positive (**0.85**), negative (**0.78**), or neutral (**0.71**).

Key Findings

- ❖ **Majority** of reviews are **positive**, highlighting platform usability and service quality.
- ❖ **Negative** reviews often focus on customer support and **food**.
- ❖ **Neutral** reviews are mostly **informative** with no clear emotional tone.



INSIGHTS

Results from Analytical Models

Churn Prediction Model

- ❖ Identified **10,701 at-risk users**.
- ❖ Model performance: **Precision = 81%**, **Recall = 89%**.

RFM Segmentation

- ❖ Customers were divided into **8 segments** based on their Recency, Frequency and Monetary values.
- ❖ Focus the strategies on customer value.

Sentiment Analysis

- ❖ Negative, neutral and positive comments were distinguished.
- ❖ Enables **personalized communication strategies**.

Customer Base Overview

Customer Value and Baseline Scenario

- ❖ Active users (2024/25 season): **106,139**
- ❖ Average Order Value (AOV): **€109.49**
- ❖ Average Orders per User: **2.37**
- ❖ Average Seasonal Spending per User: **€223.46**
- ❖ Observed churn rate (from 2023/24 to 2024/25): **70.36%**

The high churn rate value confirms the **urgent need for a retention campaign**.

DEPLOYMENT

Campaign Design

When is the campaign triggered?

- ❖ At the **start of the new season** (peak churn risk)
- ❖ Monthly refresh of churn scores for updated targeting

Who is the target audience?

- ❖ About **10,701** predicted churners

Promotion:

- ❖ **Voucher** = **20%** of AOV $\approx$ €21.90 with expected redemption rate = **10%**.

Campaign Design

We created personalized promotional messages based on customer sentiment for a consistent tone:

- ❖ **Negative Sentiment (Empathetic)**

"We sincerely regret that your past experience did not meet expectations. To demonstrate our commitment to improving, we offer you a 20% voucher for your next order. We would be pleased to welcome you back and rebuild your trust"

- ❖ **Neutral Sentiment (Encouraging)**

"A new season has just begun, and we invite you to continue your journey with us. As a token of encouragement, we provide you with a 20% voucher for your next order. We look forward to seeing you again soon."

- ❖ **Positive Sentiment (Rewarding loyalty)**

"We greatly appreciate your loyalty and the positive experiences you have shared with us. As recognition, we are pleased to offer you a 20% voucher for your next order. Thank you for being part of our community."

Campaign Economics

Campaign Costs

- ❖ Outreach = **€2,140** (€0.20 per user)
 - ❖ Voucher = **€23,431**
 - ❖ Total = **€25,571**

Incremental Retention (uplift scenarios):

- ❖ Conservative (+5%): ~**433** users saved
- ❖ Realistic (+15%): ~**1,299** users saved
- ❖ Optimistic (+25%): ~**2,165** users saved

Incremental Revenue (per user = €223.46):

- ❖ Conservative = **€96,757**
 - ❖ Realistic = **€290,275**
 - ❖ Optimistic = **€483,791**

Cost-Benefit Balance

Net Gains

- ❖ Conservative = **€71,186**
- ❖ Realistic = **€264,703**
- ❖ Optimistic = **€458,220**

Cost per Saved User

- ❖ Conservative = **€59.1**
- ❖ Realistic = **€19.7**
- ❖ Optimistic = **€11.8**

Error Costs

- ❖ **False Positives** (2,044):
wasted outreach/vouchers ≈ **€4.9k.**
- ❖ **False Negatives** (1,043):
lost revenue ≈ **€233k.**

Tailored Campaign Vouchers Based on RFM

Up to now, our campaign assumed a uniform 20% discount for all customers. Now, explore an **adaptive voucher strategy** to tailor incentives based on **RFM**.

SEGMENT	GOAL	VOUCHER VALUE
Champions	Low incentive, already loyal	10%
Loyal	Moderate incentive, consolidate loyalty	15%
Big Spenders	Moderate incentive, stimulate frequency	15%
Frequent Buyers	Moderate incentive, stimulate spending	15%
New / Promising	Incentive to foster loyalty	20%
At Risk	High incentive, reactivate	25%
Lost	Very high incentive, recover	30%
Low Value / Occasional	Base incentive, encourage purchases	20%

CONCLUSIONS

FINAL ASSESSMENT AND FUTURE OUTLOOK:

- ❖ The campaign shows **strong profitability** across all uplift scenarios.
- ❖ Expected results: **ROI highly positive** in each scenario (conservative: **+278.5%**, realistic: **+1035.1%**, optimistic **+1792%**)
- ❖ Cost-benefit balance: outreach/voucher costs are **minimal** vs. revenue **potential** which is highly influenced by FNs, which are the main limitation.

FUTURE WORK

- ❖ Improve **recall of churn model** to minimize lost revenue.
- ❖ Conduct **A/B testing** for personalized messaging effectiveness.
- ❖ Try to take advantage of **weekly seasonality** to develop more specific marketing strategies.

THANKS FOR THE ATTENTION
